Laura Xing

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Education

Columbia University, New York, NY *Expected 2027*
Master of Science in Mechanical Engineering
Research Track, with a concentration in Robotics and Control
GPA: 4.00

Johns Hopkins University, Baltimore, MD *May 2025*
Bachelor of Science in Biomedical Engineering
Focus Area in Imaging and Medical Devices
GPA: 3.78

Relevant Coursework: Principles of Design of BME Instrumentation, Electronic Properties of Materials, Microfabrication, MEMS Systems, Robot Sensors/Actuators, PCB Design and Microcontroller Programming, Applied Bioelectrical Engineering, Nanoscale Actuation/Sensing

Research Interests

My research interests lie in tactile sensing, wearable electronics, and robotic control, with a focus on designing assistive devices and environment-aware robots. I am particularly interested in integrating sensing modalities, embedded systems, and signal processing to enhance physical interaction, perception, and control in robotics and healthcare applications.

Publications & Workshop Papers

* denotes equal contribution

Xing, L.*, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Single-Pixel Tactile Skin via Compressive Sampling." *Communications Engineering*, vol. 5, article 154, 2026. doi:10.1038/s44172-026-00697-2.

Xing, L., Chang, E. T., Ballentine, P., Kymissis, I., and Ciocarlie, M. "LEAPS: Large-area Environment-aware PVDF Skin for Rapid Dynamic Sensing." *IEEE/RSJ IROS Workshop on Scalable Tactile Sensing for Dexterous Manipulation*, 2026. Accepted workshop submission (poster presentation; non-archival).

Posters

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Xing, L., Chang, E. T., Ballentine, P., Kymissis, I., and Ciocarlie, M. "LEAPS: Large-area Environment-aware PVDF Skin for Rapid Dynamic Sensing." *IEEE/RSJ IROS Workshop on Scalable Tactile Sensing for Dexterous Manipulation*, 2026. Accepted poster.

Xing, L.*, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Scalable Distributed Tactile Skin via Compressive Sampling." *Northeast Robotics Colloquium (NERC)*, 2025.

Xing, L.*, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Scalable Distributed Tactile Skin via Compressive Sampling." *Princeton Robotics Inaugural Symposium*, Princeton University, April 2026.

Research Experience

ROAM Lab **August 2025 – Present**
Columbia University
Research Assistant, advisor: Matei Ciocarlie

- **LEAPS:** Develop flexible, large-area PVDF robotic skin for dynamic contact detection, using custom low-noise amplification electronics with sampling rates up to 4 kHz.
- Characterize sensitivity, noise, dynamic range, and temporal response through feedback resistor and capacitor configurations; explore wireless communication and closed-loop safety control as future integration directions.
- **SpikeATac:** Advance fabrication methods for soft robot fingers, including elastomer casting, to improve signal consistency, manufacturability, and component reusability.

Neuroengineering & Biomedical Instrumentation Lab **Oct. 2023 – Aug. 2025**
Johns Hopkins Whiting School of Engineering
Research Assistant, advisor: Nitish Thakor

- Designed and implemented SPTS, a flexible, piezoresistive, ATtiny412-based tactile skin that addresses wiring and readout challenges in large-area tactile sensing.
- Applied hardware-embedded compressive sampling to combine sensor outputs into a shared measurement channel and reconstruct spatial pressure maps from limited measurements.
- Demonstrated $\geq$98% object classification accuracy using 20 measurements (3,500 effective frames/s) in a 10×10 sensor array; results published in *Communications Engineering* (2026).

Division of Infectious Diseases **August 2022 – May 2023**
Johns Hopkins School of Medicine
Research Assistant, advisor: Andrea Cox

- Investigated the role of LRRK2 protein in modulating NLRP3 inflammasome activation, including its involvement in inflammasome assembly and caspase-1 activation.
- Performed laboratory techniques including cell counting, cell passaging, preparation of whole-cell lysate, and western blotting.

Teaching & Service

Teaching Assistant *January 2025*
PCB Design and Microcontroller Programming, Johns Hopkins University

- Assisted students with PCB design, electronics prototyping, and microcontroller programming.
- Provided support with debugging and implementation during course and lab activities.

Student Mentor *July 2024 – August 2024*
JHU Immersive Summer Program for Education, Enrichment, and Distinction (ISPEED) in Biomedical Engineering

- Organized and hosted meeting sessions to introduce high-school students to subjects in biomedical engineering and advised them on college applications.
- Supported and developed continuous engagement with student mentees.

Skills

- **Programming:** Java, C++, Python, MATLAB
- **Microfabrication:** Photolithography, etching, thin film deposition
- **Software:** CAD, KiCad, COMSOL Multiphysics, LTspice
- **Prototyping/Instrumentation:** Circuit design, microcontroller programming, I2C/SPI, wireless communication, soldering

Hobbies

Singing (mezzo-soprano), opera, cooking, running, and visiting art museums.